

Jamal Ghobrial

AI Engineer

ghobrialjamal@gmail.com | +201501705943 | kaggle.com/jamalghobrial | linkedin.com/in/jamal-ghobrial |

Summary

Results-driven AI Engineer with a strong foundation in Machine Learning, Deep Learning, and Generative AI, backed by a BSc in Bioinformatics (GPA 3.61) and a recognized graduation project. Experienced across the full AI development lifecycle — from data engineering and model training to deploying LLM-powered agents, multi-modal search systems, and RAG pipelines in production environments. Proven ability to collaborate in agile, cloud-based teams (Azure, Docker, Git/GitHub) and deliver scalable intelligent systems that solve real-world problems. Competitive Kaggle practitioner (top 29%) with certifications from NVIDIA, Microsoft, and WorldQuant University.

Experience

AI Engineer Intern

Feb 2026 - May 2026

Areeb Technology, EG

- Engineered and deployed comprehensive AI solutions including **multi-agent content generation pipelines**, **Learning-to-Rank (LTR)** search engines, and **multi-modal e-commerce discovery platforms** (text, visual, and audio search).
- Architected end-to-end AI pipelines integrating **LLMs**, **RAG frameworks**, and **vector databases** into production environments, leveraging **Docker**, **Modal**, and **Streamlit** for scalable, serverless deployment.
- Collaborated in an agile team environment leveraging **Microsoft Azure** infrastructure and **Git/GitHub** for robust version control, streamlining code reviews through detailed pull request (PR) reports.

AI Engineer Intern

Oct 2025 - Jan 2026

National Telecommunications Institute (NTI), EG

- Gained end-to-end experience building and evaluating **Machine Learning**, **Deep Learning**, **Computer Vision**, and **NLP** models using Python, TensorFlow, and PyTorch, with emphasis on structured data pipelines and performance benchmarking.
- Developed a multi-stage, **multi-modal AI evaluation system** integrating LLM-based grading, **RAG retrieval**, **audio/visual processing**, and **automated PDF/PPTX analysis** to generate structured scoring and performance assessments for competition submissions.

Microsoft Machine Learning Engineer Intern

Jun 2025 - Dec 2025

DEPI, EG

- Developed advanced AI/ML models and neural networks for complex tasks like classification, prediction, and clustering.
- Applied AI principles to build innovative, intelligent systems for task automation and operational improvement.

Machine Learning Intern

Jul 2023 - Oct 2023

EME Initiative(Egypt Makes Electronics) – ITIDA, EG

- Developed and optimized a defect prediction model using Python (Scikit-learn, XGBoost), achieving 85% accuracy by engineering robust datasets, applying dimensionality reduction, and performing rigorous hyperparameter tuning to boost model performance by 40%.
- Demonstrated advanced modeling skills by developing complex algorithms to achieve top 29% and 33% rankings in competitive Kaggle challenges.

Technical Skills

Programming Languages: Python (Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow), Go, C++.

Machine Learning Fundamentals: Machine Learning Algorithms, Data Modeling, Feature Engineering, Learning-to-Rank (LTR), LightGBM, ML Pipeline.

LLM Architectures: Transformers, Encoder-Decoder (BERT/T5), Decoder-Only (GPT/Llama), Attention Mechanisms, Tokenization (BPE/WordPiece).

Generative AI & RAG: Prompt Engineering, Retrieval Augmented Generation (RAG), Vector Databases (Qdrant), Mixture-of-Agents (MoA), LibreChat, Semantic Search.

AI Frameworks: Hugging Face Transformers, SentenceTransformers, PyTorch, TensorFlow, OpenAI API, Neural Network Architectures.

Cloud & DevOps: AWS SageMaker, Microsoft Azure, Docker, Modal, Git/GitHub.

Others: Image Processing, Data Visualization (Streamlit, Plotly Dash), ETL Processes, SQL, Tantivy.

Soft Skills

• Problem-Solving • Analytical Skills • Continuous Learning • Initiative • Results-Oriented • Self-Management • Self-Motivation • Adaptability

Projects

Semantrix: Learning-to-Rank Search Engine

- Engineered a two-stage search engine combining **Tantivy** for fast retrieval and **LightGBM LambdaRank** for relevance ranking on MS MARCO.
- Built a serverless **Streamlit** UI backed by **Modal** for fast inference and real-time visualization of semantic similarities.

OmniFind: Multi-Modal Search App

- Developed a full-stack product discovery web app using **Express**, **React**, **Node.js**, and **MongoDB**, containerized via **Docker**.
- Implemented multi-modal search, including **semantic text search**, **DINO/Qdrant-powered visual search**, and **audio transcription** using Python workers.

Intelligent Content Creator Agent

- Designed a **Mixture-of-Agents (MoA)** pipeline using **LibreChat** to autonomously generate platform-adapted content.
- Integrated **Web Search (Tavily)**, **RAG**, and multiple LLMs (**Claude**, **GPT-4**, **Gemini**) to ensure brand voice consistency and format compliance.

Digitopia Assistant Judge (NTI Internship)

- Designed and implemented a **multi-modal AI evaluation system** integrating audio processing, computer vision, **PDF/PPTX parsing**, and **Retrieval-Augmented Generation (RAG)** to automatically assess competition submissions with structured scoring logic.
- Created an **LLM-powered grading agent** using **Python**, **SQLite**, and **custom pipelines** to generate automated scoring, reasoning, and evaluation reports, improving assessment accuracy and reducing manual review time.

Courses & Certifications

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| • Getting Started with Deep Learning , NVIDIA | Dec 2025 |
| • Applied Data Science, WorldQuant University | Jul 2023 |
| • Data Analysis Professional Nanodegree, Udacity | Dec 2022 |

Education

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| Faculty of Computers and Information, BSc in Bioinformatics
Assiut University | Jul 2023 |
| • GPA: 3.61/4.0 | |
| • Graduation Project: Recognized for excellence. | |

Additional Information

Military Status

- Completed (Jun 2025)